Chapter 4: Model checking in the Cox regression model — Exercises

STA310 Survival Analysis (Master's in Applied Biostatistics)

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Chapter 4 covers the diagnostics used to assess whether a fitted Cox model is adequate: the various residuals (Cox–Snell, martingale, deviance, Schoenfeld, score), their use in checking overall fit and the functional form of covariates, the detection of influential observations, and — most importantly — checking the proportional hazards assumption.

1 Residuals for the Cox regression model (Book §4.1)

1.1 Cox–Snell residuals

Type: Concept & notation

The Cox–Snell residual for the ith individual is

$$r_{Ci} = \exp(\hat{\beta}' x_i)\, \hat{H}_0(t_i).$$

(a) Express r_{Ci} in terms of the estimated cumulative hazard (or survivor function) of the ith individual.
(b) If the fitted model is correct, what distribution do the Cox–Snell residuals approximately follow? State its mean, and say whether it is symmetric.
(c) What happens to the residual of a right-censored individual, and how does this complicate the use of the residuals for checking?

1.2 Martingale residuals

Type: Concept & notation

Builds on: Chapter 3

The martingale residual is $r_{Mi} = \delta_i - r_{Ci}$, where δ_i is the event indicator.

(a) Interpret r_{Mi} as the difference between two quantities.
(b) What is the range of possible values, and what sign do censored observations always have?
(c) What property, shared with residuals from linear regression, do the martingale residuals have?

1.3 Deviance residuals

Type: Concept & notation

(a) Why were deviance residuals introduced, given that martingale residuals already existed?
(b) In broad terms, how do they relate to martingale residuals?
(c) Do deviance residuals sum to zero?

1.4 Schoenfeld residuals

Type: Concept & notation

Builds on: Chapter 3

The Schoenfeld residual for individual i and explanatory variable X_j is $r_{Sji} = \delta_i \left(x_{ji} - \hat{a}_{ji}\right)$, where $\hat{a}_{ji}$ is a risk-set-weighted average of X_j among the individuals at risk at time t_i.

(a) How many Schoenfeld residuals are there for each individual, compared with the Cox–Snell or martingale residuals?
(b) Why are the residuals zero for censored observations, and what does $\hat{a}_{ji}$ represent?
(c) For what specific purpose are the (scaled) Schoenfeld residuals especially useful?

2 Assessing model fit (Book §4.2)

2.1 The cumulative hazard plot of the residuals

Type: Interpret figure

A common diagnostic is a plot of $\hat{H}(r_{Ci}) = -\log \hat{S}(r_{Ci})$ — the estimated cumulative hazard of the Cox–Snell residuals — against the residual values r_{Ci}.

(a) What is being plotted, and why does a straight line through the origin with unit slope indicate a good fit?
(b) You obtain a plot that curves systematically away from that line. What do you conclude?

(c) Why is this plot regarded as insensitive — not to be relied on by itself?

2.2 Checking the functional form of covariates

Type: Diagnostics & model critique

(a) Describe how martingale residuals from the null model can be used to check the functional form of a continuous covariate.
(b) What shape in the plot indicates that a linear term is adequate, and what would indicate that a transformation is needed?
(c) In the myeloma example, the smoothed martingale plot against Bun was roughly straight, but the plot against $\log(\text{Bun})$ was curved. What does this suggest about how Bun should enter the model?

2.3 Identifying poorly fitted observations

Type: Interpret figure

Two plots are used to look for outliers: an index plot of the deviance residuals, and a plot of the deviance residuals against the risk score.

(a) What does a large positive (or large negative) deviance residual indicate about an individual?
(b) Why are deviance residuals preferred over martingale residuals for these plots?
(c) What is the value of plotting the deviance residuals against the risk score rather than just making an index plot?

3 Influential observations (Book §4.3)

3.1 Delta-betas and influence

Type: Concept & notation

Builds on: Chapter 3

(a) What is a delta-beta, $\Delta_i\hat{\beta}_j$, and what does a large absolute value indicate?
(b) On which residual are the delta-betas based?
(c) How does influence on a single parameter estimate differ from the influence summarised by the likelihood displacement LD_i or the diagnostic $|l_{\max}|$?

4 Checking proportional hazards (Book §4.4)

4.1 The log-cumulative hazard plot

Type: Assumptions

Builds on: Chapters 1–3

For the Cox model, $\log H_i(t) = \beta' x_i + \log H_0(t)$.

(a) Explain why, under proportional hazards, the log-cumulative hazard curves for different groups should be parallel.
(b) Describe how the plot is constructed when the covariate of interest is continuous.

(c) What are the main limitations of this plot?

4.2 Schoenfeld residuals and the zph test

Type: Interpret R output

A Cox model with three variables (`treat`, `age`, `sex`) is fitted, and the proportional hazards assumption is tested. The output is:

```
           chisq df     p
treat      9.84  1 0.002
age        1.32  1 0.25
sex        0.66  1 0.42
GLOBAL    11.82  3 0.008
```

(a) What hypothesis does each row test?
(b) Which variable(s) show evidence against proportional hazards?
(c) What is the "GLOBAL" test, and what do you conclude overall?

4.3 What to do when proportional hazards fails

Type: What if …?

Suppose the zph test shows a treatment effect that decreases over time — a strong early benefit that wanes.

(a) What does it mean, substantively, that the treatment effect is time-dependent?
(b) Describe one model-based way (from §4.4.4) to capture this, and state the key added term.
(c) What does a test of that added term's coefficient amount to?

5 Putting it together

5.1 A misleading diagnostic conclusion

Type: What's wrong here?

A researcher reports: "I checked my Cox model by plotting the Cox–Snell residuals against survival time; they scattered roughly symmetrically about zero, so the model fits well."

Explain what is wrong with this statement.

6 Mini-lab: checking a Cox model in R

6.1 Detecting non-proportional hazards

Type: Mini-lab

Run the code below. It simulates 400 patients in two treatment groups. The control group has a constant hazard, but the treated group has a hazard that *increases* over time, so the treatment effect is not constant. A Cox model is fitted and the proportional hazards assumption is tested.

```r
set.seed(731)
n <- 400
treat <- rep(c(0, 1), each = n/2)
t0 <- rexp(n/2, rate = 0.05)                    # control: constant hazard
t1 <- rweibull(n/2, shape = 2, scale = 1/0.07)  # treated: increasing hazard
t <- c(t0, t1)
cens <- runif(n, 0, 60)
obs <- pmin(t, cens)
status <- as.numeric(t <= cens)
d <- data.frame(time = obs, status = status, treat = treat)

library(survival)
fit <- coxph(Surv(time, status) ~ treat, data = d)
summary(fit)
```

```
Call:
coxph(formula = Surv(time, status) ~ treat, data = d)

  n= 400, number of events= 288

        coef exp(coef) se(coef)     z Pr(>|z|)
treat 0.5322    1.7027   0.1281 4.156 3.24e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

      exp(coef) exp(-coef) lower .95 upper .95
treat     1.703     0.5873     1.325     2.188

Concordance= 0.508  (se = 0.018 )
Likelihood ratio test= 17.56  on 1 df,   p=3e-05
Wald test            = 17.27  on 1 df,   p=3e-05
Score (logrank) test = 17.62  on 1 df,   p=3e-05
```

```r
cox.zph(fit)
```

```
       chisq df       p
treat   63.6  1 1.5e-15
GLOBAL  63.6  1 1.5e-15
```

(a) From `summary(fit)`, report the estimated hazard ratio for `treat` and its p-value. What would a naive interpretation conclude?

(b) Report the test statistic and p-value from `cox.zph(fit)`. What does it tell you?

(c) Explain why the hazard ratio from (a) is misleading, and say what you should do next.